

BACEM EL MANAI

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PROFESSIONAL SUMMARY

Cybersecurity-focused developer who got into this field by breaking things apart to understand how they work, and stayed for the puzzle-solving. Spends free time reverse engineering binaries and picking apart malware samples on TryHackMe, HackTheBox, and LetsDefend, and brings that same "assume nothing, verify everything" mindset into every application built. Has shipped production systems for real clients while treating security as a first-class requirement, not an afterthought, from MFA and RBAC design to penetration testing before deployment. Comfortable moving between offense and defense: exploiting memory corruption bugs and writing exploit scripts one day, hardening authentication flows and reviewing code for vulnerabilities the next. Tools of choice include Ghidra, Metasploit, and Burp Suite on the security side, React and Flask on the build side.

TECHNICAL SKILLS

Languages: Python, JavaScript, C, C++, C#, Java, Assembly, Bash, SQL, PHP

Frontend: React.js, Next.js, HTML5, CSS3, SCSS

Backend: Flask, Node.js, PHP, Express

Databases: MySQL, MariaDB, PostgreSQL, MongoDB

Security Tools: Metasploit, Nmap, Burp Suite, Wireshark, Ghidra, IDA Free, radare2, gdb, ffuf, enum4linux, sqlmap, Wazuh, Suricata, nftables

DevOps: Docker, Git, GitHub, AlwaysData VPS, AWS EC2, AWS, OpenTofu, Terraform, Kubernetes

Design: Figma

IDEs & Editors: VS Code, Visual Studio Community, Visual Studio Insiders, Thonny, Code::Blocks, nano, Vim

PROFESSIONAL EXPERIENCE

Secure Software Developer

February 2026 - May 2026

Zimys • Ben Arous, Tunisia

- Design and develop secure software solutions with security-first architecture and threat modeling
- Implement robust authentication and authorization systems with multi-factor authentication and role-based access control
- Conduct security code reviews and penetration testing to identify and remediate vulnerabilities before production deployment
- Build secure RESTful APIs and web applications with Python, Flask, and modern security frameworks

Web Developer

October 2025 - Present

Engineers Spark

- Develop and maintain 3+ production web applications serving 500+ monthly active users using React.js, Flask, and JavaScript
- Build scalable backend systems with Flask and Python that process 10,000+ API requests daily
- Create responsive frontend interfaces and integrate with backend services using REST APIs
- Collaborate with cross-functional teams to deliver high-quality web solutions with 99.8% uptime
- Implement security best practices reducing vulnerabilities by 75% through code optimization and security patches
- Work with Docker for containerized development and Git for version control, improving deployment efficiency by 40%

Secure Software Development Intern

January 5 - January 31, 2026

Zimys • Ben Arous, Tunisia

- Architected and deployed MediCare Health Platform - a secure, role-based web application for digitizing diagnostic report distribution across doctors, patients, and medical centers in Tunisia
- Designed 7-table MariaDB schema with foreign key constraints, cascading deletes, ENUM types, and optimized indexes supporting 5,000+ monthly transactions
- Engineered Flask 3.0 backend with three distinct portals featuring MFA authentication (bcrypt + 5-minute OTP expiry), RBAC controls, and parameterized queries preventing all SQL injection attempts
- Implemented secure file handling system with 16MB size limits, MIME validation, UUID-based filenames, and organized storage - processing PDF/DOCX/CSV/XLSX/DICOM formats with zero path traversal vulnerabilities
- Conducted comprehensive penetration testing using sqlmap, Burp Suite, nmap, and Wireshark - achieving 99.21% availability under DoS simulation and zero critical vulnerabilities in production
- Deployed to AlwaysData VPS (Paris) with Python 3.14 environment, achieving 100% uptime and reducing report delivery time from 24-72 hours to under 1 minute

Web Developer (Freelance)

June 2025 - September 2025

High Gloss Design

- Developed Kitchen Price Editor & Quote Generator web application reducing quotation preparation time by 85%
- Implemented shopping cart functionality and auto-search features increasing conversion rates by 25%
- Integrated PDF quote generation system using jsPDF automating 100+ client quotations monthly
- Built secure authentication system with SHA-256 hashing and brute-force protection preventing 15+ attempted breaches
- Implemented timing attack prevention using Web Crypto API enhancing overall application security posture

Software Developer (Freelance)

January 2025 - May 2025

FATMA CHIMIE

- Designed and developed complete secure inventory management system for industrial chemical company processing 5,000+ transactions monthly
- Implemented comprehensive security features including user authentication and data encryption protecting sensitive chemical formulas
- Built responsive frontend interface with React.js, HTML5, and CSS3 improving user workflow efficiency by 60%
- Developed robust backend architecture with database integration using Node.js, PHP, and SQL
- Deployed production system with real-time inventory tracking and reporting capabilities reducing inventory errors by 45%

PROJECTS

Supply Chain Security Analyzer - Polyglot Microservice SCA Platform

2026

Lead Developer • Personal Project • Software Supply Chain Security

- Architected a multi-language software composition analysis tool spanning Go, Rust, and C++, communicating via gRPC/Protocol Buffers across four independent microservices
- Built a Go orchestrator parsing npm package.json/package-lock.json (both v1 and v2/v3 lockfile formats), constructing a dependency graph with blast-radius and depth analysis, and querying the OSV vulnerability database with batched requests, retry/backoff, and CVE detail enrichment
- Developed a Rust typosquatting detection service using Levenshtein distance, Jaro-Winkler phonetic similarity, and Unicode homograph normalization against popular npm packages
- Developed a Rust secret-scanning service combining regex signature detection (AWS/GitHub/Stripe keys, JWTs, private keys) with Shannon entropy analysis, scanning both the working tree and full Git history via libgit2 with redaction enforced at the protobuf schema level
- Built a C++ CI/CD pipeline analyzer using RapidYAML to detect GitHub Actions vulnerabilities including curl-pipe-to-shell execution, unpinned action references, and the pull_request_target privilege-escalation pattern
- Containerized all four services with multi-stage Docker builds and deployed the complete stack on Kubernetes (Deployments, Services, ConfigMap, and a Job for orchestrated scans), validated end-to-end on a local kind cluster
- Repository: github.com/becem69/Supply-Chain-Security-Analyzer

SOC Home Lab with Wazuh - Dockerized Detection & Response Pipeline

2026

Personal Project • Security Operations & Detection Engineering

- Architected a full SOC detection pipeline from scratch, running Wazuh (SIEM/XDR), a monitored victim host, Suricata IDS, and an attack target entirely in Docker containers instead of full VMs to fit low-end hardware constraints
- Deployed and debugged single-node Wazuh Manager, Indexer, and Dashboard, diagnosing an OpenSearch OutOfMemoryError down to a misconfigured off-heap memory pool rather than the JVM heap itself
- Configured a monitored Ubuntu victim host with a version-pinned Wazuh agent (FIM, rootcheck, syscollector) and deployed OWASP Juice Shop as a live, deliberately vulnerable attack target
- Stood up Suricata as a host-level network IDS sniffing the custom Docker bridge interface, with custom detection rules for Juice Shop attack traffic
- Integrated the Wazuh community detection ruleset with automated updates, plus a full VirusTotal integration for File Integrity Monitoring, verified end-to-end with an EICAR test file (65/67 engines flagged, alert auto-generated with full verdict)
- Implemented network-level enforcement with host nftables rules hooked into Docker's DOCKER-USER forward chain (requiring br_netfilter), isolating the monitored host from core SOC infrastructure with logged, counter-tracked block rules
- Hardened persistence across reboots via systemd services, module autoloading, and systemctl configuration for the full firewall and detection stack
- Repository: github.com/becem69/soc-homelab-wazuh

Olive Factory Inventory Management System

2026

Lead Developer • Commissioned Client Project (Sold Application)

- Built and delivered a full-stack inventory management platform for an olive oil production facility using Python, Flask, and MySQL
- Designed real-time stock tracking with automatic quantity updates on purchases and sales, including low-stock alerting

- Implemented financial dashboard surfacing revenue, spending, and net profit over rolling 30-day windows
- Developed supplier and client relationship management with searchable partner directories and transaction history
- Engineered container logistics tracking system computing net inventory by reconciling "Sent" vs. "Received" movements
- Delivered production-ready AdminLTE/Bootstrap interface and deployed to AlwaysData hosting; application sold to client for commercial use
- Repository: github.com/becem69/olive-factory-application

CyberSpark CTF - Reverse Engineering Challenges Author

2026

Author & Challenge Designer • CyberSpark CTF

- Designed and implemented a progressive reverse engineering track simulating real-world escalation from basic string analysis to advanced symbolic execution
- Developed 6 multi-platform challenges covering C binaries, Java bytecode, PyInstaller-packed Python, Z3-based logic, Z3-based obfuscated logic, and JavaScript runtime verification
- Technical focus areas: static/dynamic binary analysis, PyInstaller unpacking, JVM bytecode decompilation, SMT solving with Z3 theorem prover, and client-side JavaScript debugging
- Challenges included: top_sec (C string analysis), unsolvable_challenge.java (XOR decryption), easy_snake (Python unpacking), Z3_obfuscation.py (symbolic execution), Z3.py (Z3-logic program), and the_vault.js (bitwise operations)
- Repository: github.com/becem69/CyberSpark-reverse-engineering-challenges

MediCare Health Platform - Secure Healthcare Report Distribution System

2026

Lead Developer • Zimys Internship Project

- Designed and implemented full-stack Flask 3.0/MariaDB application with three role-based portals (doctors, patients, medical centers) serving Tunisia's healthcare sector
- Engineered security-first architecture featuring bcrypt password hashing, 5-minute expiring OTP via email, RBAC controls, and parameterized queries blocking 100% of SQL injection attempts
- Built secure file handling system supporting PDF/DOCX/CSV/XLSX/DICOM with 16MB limits, MIME validation, UUID filenames, and zero path traversal vulnerabilities
- Conducted comprehensive penetration testing with sqlmap (SQLi), Burp Suite (XSS), nmap (network scanning), and Wireshark (traffic analysis), achieving zero critical vulnerabilities
- Deployed to AlwaysData VPS (Paris) with Python 3.14 environment; achieved 99.21% availability under DoS simulation and reduced report delivery time from 24-72 hours to under 1 minute
- Live at: bacem69.alwaysdata.net • Repository: github.com/becem69/MediCare-Health-Platform-project

WireGuard VPN Server Configuration - Secure Tunneling Infrastructure

2026

Network Security Engineer

- Designed and deployed end-to-end WireGuard VPN infrastructure on Ubuntu server with stealth UDP/53 configuration to bypass restrictive firewalls
- Configured IP forwarding, iptables NAT rules, and systemd service for persistent operation on cloud VPS environments
- Implemented secure key exchange between server and client using wg genkey, enforced strict peer routing with AllowedIPs, and enabled PersistentKeepalive for NAT traversal
- Optimized for offensive security use cases: traffic masquerading, DNS-port obfuscation, and minimal attack surface
- Repository: github.com/becem69/VPN-server-config

cook.bin - ELF Binary Reverse Engineering Challenge

2026

Author & Reverse Engineer • Ramadhan CTF (ISET'COM)

- Analyzed stripped 64-bit PIE ELF binary using static analysis techniques (file, strings, objdump) to locate XOR-encoded flag data in .rodata section
- Identified XOR obfuscation pattern and reverse-engineered key (0xDE) by analyzing null-byte termination in encoded sequence at offset 0x2090
- Developed Python decryption script to recover hidden flag: becem69_is_4lw4ys_h3r3
- Documented complete solution walkthrough including anti-debugging bypass (/proc/self/status inspection) and decoy flag analysis
- Repository: github.com/becem69/Ramadhan-CTF-cook-task-writeup

rev_this_#.bin - .NET Self-Contained Binary Reverse Engineering

2026

Author & Reverse Engineer • Ramadhan CTF (ISET'COM)

- Reverse engineered .NET 6 self-contained ELF binary containing hidden managed assembly (revproj.dll) embedded within Linux executable
- Located .NET bundle header using magic byte signature (0x8B1202B96A612038) and extracted 10KB DLL payload at offset 0xa49730
- Analyzed .NET metadata streams (#US, #Blob, FieldRVA) to recover XOR key (48 bytes), encrypted flag halves (24 bytes each), and SHA-256 verification hash
- Implemented XOR decryption in Python to reconstruct full flag: jus7_r3m3mb3r_n0_c4p_wh3n_y0u_sp3ll_7h3m_becem69
- Verified solution by matching SHA-256 hash against TARGET_HASH and confirming "Access Granted!" response from original binary
- Repository: github.com/becem69/Ramadhan-CTF-rev-this-sharp-challenge-writeup

PORTVIGIL - Advanced Multi-Threaded Port Scanner

2025

Author & Security Tool Developer

- Designed and implemented high-performance, CLI-based network port scanner in Python for penetration testing
- Engineered concurrent scanning using ThreadPoolExecutor, achieving scan rates over 1,000 ports/second
- Integrated intelligent service detection (SSH, HTTP, RDP, MySQL) with security recommendations
- Supported flexible port input (ranges, lists, presets) and JSON export for reporting
- Built with robust error handling, DNS resolution, and user-friendly CLI using argparse
- GitHub repository with 50+ stars and contributions from 5 community developers
- Repository: <https://github.com/becem69/PORTVIGIL>

CYBERSECURITY EXPERIENCE

Competitive Cybersecurity: CTF Player

2024 - Present

Specialization: Reverse Engineering & Binary Exploitation (Pwn)

- Reverse engineer x86/x64 and ARM binaries using Ghidra, IDA Free, and radare2
- Analyze and write Assembly (x86, x86_64) to understand program logic and bypass protections
- Exploit memory corruption bugs: stack/heap overflows, ROP, format strings, use-after-free
- Develop custom exploit scripts in Python and C for reliable shellcode delivery

Malware Analysis & Reverse Engineering Challenges

2025 - 2026

Platforms: TryHackMe, HackTheBox, LetsDefend

- Solved 8 malware analysis and reverse engineering challenges spanning static/dynamic analysis, obfuscation defeat, and RAT behavioral analysis

- TryHackMe: Grand Larceny Auto (medium), Directory (hard), REloaded (hard), Basic Malware RE (medium)
- HackTheBox: ShadowMonarch (hard), Foreigner (hard), Subatomic (medium), Antarctica (medium)
- LetsDefend: Obfuscated RAT (hard), SOC-style incident triage and de-obfuscation of a remote access trojan

Linux Systems and Penetration Testing

2020 - Present

Primary Operating System: Kali Linux

- Conducted 15+ penetration tests for small businesses identifying and remediating critical vulnerabilities
- Network security analysis with Nmap, Wireshark, and Burp Suite
- Vulnerability assessment and digital forensics
- Developed 20+ Bash scripts automating security tasks and system hardening procedures
- Metasploit framework for security testing
- System hardening, malware analysis, and Linux kernel customization

CERTIFICATIONS

Intro to AI and Gen AI for Nomu AI Ghurair TechUp Explorer

May 2026

Abdulla Al Ghurair Foundation - TechUp Explorer • Issued by Udacity (Part of Accenture)

- Verified Certificate of Course Completion • Recipient: Bacem El Manai
- Signed by Kai Roemmelt, CEO of Udacity • Date Awarded: May 22, 2026

Python Developer Certification

December 2025

freeCodeCamp.org • 300 Hours

- Certificate ID: fccee0cf537-0bdf-4994-a22b-7b7762e4231f/python-v9

Getting Started with OpenTofu (LFEL1009)

December 2025

The Linux Foundation

- Certificate ID: LF-eym1nzu4e3

Recon Certificate of Completion

November 2025

PentesterLab

- Certificate ID: PTLR2977

OSINT (Open Source Intelligence)

October 2025

Cybrary • 1 CEU/CPE • 2 Hours

- Certificate ID: CC-d18da919-cee5-4e43-b5d3-ab0089e211a2

Programming 100: Fundamentals

October 2025

TCM Security • 3 CEU Hours

- Certificate ID: cert_nkv3qqpx

Cybersecurity Essentials (LFC108)

October 2025

The Linux Foundation

Introduction to Cybersecurity

October 2025

Cisco Networking Academy

Linux 100: Fundamentals

September 2025

TCM Security • 2 CEU Hours

- Certificate ID: cert_z8ng8xtd

EDUCATION

Higher Institute of Technological Studies in Communications (ISET'COM) *Ariana, Tunisia*
Bachelor of Technology, Management of Information and Communication Technologies (MICT),
2025-2028
Current focus: Cybersecurity and Software Development

Xiamen University Malaysia *Selangor, Malaysia*
Software Engineering, 2024-2025

Abderrazek Hammemi High School *Tunisia*
High School Diploma, Information Technology Major, 2024

LANGUAGES

- Arabic: Native fluency
- English: Fluent
- French: Elementary